

scaleLLMflow

General LLM Workflows for Scientific Quality Scales

Proyecto MQS

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URL <https://github.com/ppmena/scaleLLMflow>

Contents

1	Package scaleLLMflow	3
1.1	Description	3
1.2	Two workflow modes	3
1.3	Credentials and provider configuration	3
2	Registry and scale helpers	3
2.1	available_scales	3
2.2	available_provider_models	3
2.3	resolve_prompt	4
2.4	audit_model_registry	4
3	Prompt and text preparation	5
3.1	build_scale_prompt	5
3.2	extract_article_text	5
3.3	extract_pdf_text	5
3.4	convert_article_markdown_llm	6
3.5	strip_references_section	6
4	Parsing and provider calls	6
4.1	parse_scale_scores	6
4.2	run_llm	7
5	Article and dataset workflows	7
5.1	run_article	7
5.2	run_dataset	8

6	Bundled scales and registry layout	8
7	Examples	8
8	Index	9

1 Package scaleLLMflow

1.1 Description

Reusable helpers for registering scientific quality scales, resolving prompts by provider and model, sending article text to Gemini, OpenAI, or Claude, validating strict JSON responses, parsing item scores, comparing them with reviewed references, and writing audit and evidence files.

The package deliberately separates three layers: the formal scale definition in metadata, the operational prompt sent to the model, and the numeric coding used for analysis. The bundled MQS and PEDro registries provide provider-generic prompts for Gemini, OpenAI, and Claude, plus selected model-specific variants.

1.2 Two workflow modes

Free mode The model response is scored and preserved for inspection without assuming that a previous answer is correct.

Reference mode A reviewed score vector is supplied and every returned item is compared with it. This mode reports agreement and mismatches while preserving the model's evidence and reasoning.

1.3 Credentials and provider configuration

Gemini credentials are read from `GEMINI_API_KEY` or `GOOGLE_GEMINI_KEY`. OpenAI credentials are read from `OPENAI_API_KEY`; `OPENAI_PROJECT_ID` may also be supplied. A project-level `.Renviron` file is convenient for interactive work in RStudio and should not be committed. Claude credentials are read from `ANTHROPIC_API_KEY` or `CLAUDE_API_KEY`.

2 Registry and scale helpers

2.1 `available_scales`

Description

Lists scales available in the prompt registry.

Arguments

registry_dir Optional registry root. The bundled registry is used when omitted.

Value

A character vector of scale identifiers.

2.2 `available_provider_models`

Description

Queries the provider's live model catalogue. This function is independent of the scale registry and does not use hard-coded model lists.

Arguments

provider `gemini`, `openai`, `claude`, or an alias. The aliases `chatgpt` and `anthropic` are accepted where applicable.

api_key Optional in-memory key. Otherwise the provider environment variables are used.

timeout Request timeout in seconds.

supported_only For Gemini, retain models supporting `generateContent`.

Value

A data frame with provider, model identifier, and display name. OpenAI uses `OPENAI_API_KEY`; Gemini uses `GEMINI_API_KEY` or `GOOGLE_GEMINI_KEY`; Claude uses `ANTHROPIC_API_KEY` or `CLAUDE_API_KEY`.

2.3 resolve_prompt

Description

Resolves the most appropriate registered prompt and its metadata. Matching proceeds from the exact model to family, numeric-version, provider-specific generic, generic, and nearest available fallbacks. The provider-specific fallback folders are named `claude-generic`, `gemini-generic`, and `openai-generic`.

Arguments

scale Scale identifier.

model Requested model identifier.

provider Optional provider used to select its generic prompt.

registry_dir Optional registry root.

Value

A resolved registry record containing prompt text, metadata, scale definition, response schema, and provenance fields.

2.4 audit_model_registry

Description

Checks every scale and model registry entry for required files, valid metadata, a complete item definition, a coherent response schema, and prompt hashes. Exact duplicate prompts are grouped and a unification plan is reported.

Arguments

registry_dir Optional registry root.

output_dir Optional directory for audit CSV reports.

apply_unifications If `TRUE`, moves redundant prompt folders to a timestamped backup after reporting the plan.

Value

An audit result with issue and success counts, duplicate groups, and proposed or applied unifications.

3 Prompt and text preparation

3.1 build_scale_prompt

Description

Combines an operational scale prompt with the article text.

Arguments

prompt_text Registered prompt text.

article_text Article text to evaluate.

Value

The complete provider prompt as a character string.

3.2 extract_article_text

Description

Reads local Markdown or text articles and optionally removes the references section. PDF input is supported through **extract_pdf_text**. For PDF articles, **conversion = "llm"** generally produces better reading order and Markdown structure, especially for multi-column articles.

Arguments

file_path Local article path.

filetype One of auto, txt, md, or pdf.

strip_references Whether to remove references before model submission.

Value

Article text.

3.3 extract_pdf_text

Description

Extracts text from a local PDF using **pdftools**. With **conversion = "llm"**, an LLM reconstructs reading order and Markdown structure; For two-column PDFs, the conversion prompt explicitly requests complete left-column then right-column reading on each page. The model is selected independently with **provider** and **model** (or **conversion_provider** and **conversion_model** in **run_article**). this generally gives better results for complex and multi-column articles. If the extracted text exceeds **max_chars** (50,000 by default), it is split at safe line boundaries, converted in several calls, and reassembled into one Markdown file.

Arguments

pdf_path Local PDF path.

strip_references Whether to remove references.

Value

Article text.

3.4 `convert_article_markdown_llm`

Description

Converts extracted article text into faithful Markdown with an LLM, reconstructing reading order, headings, and clearly identified tables. This generally improves results for complex PDF layouts.

Value

Markdown text returned by the configured provider.

3.5 `strip_references_section`

Description

Removes a detected references or bibliography section.

Value

Text with the trailing references section removed when detected.

4 Parsing and provider calls

4.1 `parse_scale_scores`

Description

Validates and parses a strict JSON scale response. Fenced JSON is normalized in a controlled way before validation. The registered metadata defines required items, allowed values, excluded items, and total-score rules.

Arguments

text Raw model response.

items Expected item identifiers.

metadata Optional resolved metadata and JSON Schema.

Value

A named numeric vector of item scores. When metadata is supplied, the response is validated against its declared schema before scores are returned.

4.2 run_llm

Description

Calls Gemini through the Interactions API, OpenAI through the Responses API, or Claude through Anthropic's Messages API. It applies rate limiting, retries transient failures with exponential backoff, records detailed errors, and returns the raw response needed for auditability.

Arguments

prompt Prompt sent to the provider.

provider gemini, openai, chatgpt, claude, or anthropic.

model Provider model identifier.

temperature Sampling temperature; zero is recommended for reproducibility.

top_p Gemini nucleus sampling parameter; ignored by OpenAI and Claude.

timeout Timeout per attempt, in seconds.

api_key Optional in-memory API key.

project_id Optional OpenAI project id.

max_retries Maximum transient-failure retries.

retry_wait_seconds Initial retry delay.

retry_backoff Retry-delay multiplier.

rate_limit_seconds Minimum delay between requests in the current R process.

Value

A provider response object containing text, request metadata, model and parameter provenance, hashes, input size, and error details when relevant.

5 Article and dataset workflows

5.1 run_article

Description

Runs one local article through a registered scale prompt. It writes raw audit output plus convenient score and evidence CSV files.

Important arguments

article_path Local article file.

scale Scale identifier.

provider Provider identifier.

model Model identifier.

output_dir Output directory.

reference_scores Named reviewed scores; activates reference validation.

validation_mode free or reference.

temperature, top_p, timeout Provider parameters.

max_retries, retry_wait_seconds, retry_backoff, rate_limit_seconds Reliability controls.

Value

A result containing parsed scores, totals, evidence, reasons, validation comparisons, provenance, and output paths.

5.2 run_dataset

Description

Processes all supported local articles in a directory and consolidates scores, evidence, audit information, and errors. A semicolon-separated reference CSV with ID and item columns activates dataset reference mode.

Value

A dataset result with per-article records and consolidated output paths.

6 Bundled scales and registry layout

The bundled registry currently contains the MQS and PEDro scales. Each provider generic folder contains exactly two files:

```
inst/scales/<scale>/<provider>-generic/  
  prompt.md  
  metadata.json
```

The metadata file is authoritative for item identifiers, allowed decisions, included items, total-score rules, and the response schema. A model-specific folder may be added only when its prompt has been separately tuned or validated.

PEDro reports all 11 items, but item 1 is excluded from the official total. MQS uses its declared 10-item scoring definition. Do not infer a total from the prompt text; use the metadata and package parser.

7 Examples

The following example is suitable for an interactive RStudio session:

```
library(scaleLLMflow)  
results <- run_dataset(  
  articles_dir = "examples/03_pedro_free/articles",  
  scale = "pedro",  
  provider = "gemini",  
  model = "gemini-3.6-flash-lite",
```



```
output_dir = "results/pedro-gemini",
validation_mode = "free",
temperature = 0,
top_p = 0.1
)
```

For Claude, configure `ANTHROPIC_API_KEY` or `CLAUDE_API_KEY` and run the bundled example. The scale and model are explicitly defined in the script:

```
Rscript examples/05_claude_free/run.R
```

Edit `scale` and `model` in that script to switch between MQS/PEDro or another Claude model.

For comparison against reviewed scores:

```
results <- run_dataset(
  articles_dir = "articles",
  scale = "mqs",
  provider = "openai",
  model = "gpt-4.1-mini",
  output_dir = "results/reference",
  reference_csv = "mqs_reference.csv",
  validation_mode = "reference"
)
```

8 Index